

Lab 3 – Architectural Justification

Hyperlocal Courier Dispatch and Tracking System

Software Engineering Laboratory

Architectural Choice

We chose **Microservices Architecture** for the Hyperlocal Courier Dispatch and Tracking System. The system is divided into independent services such as courier dispatch, route optimization, delivery verification, and tracking and notification. Each service performs a specific responsibility and communicates with other services through well-defined interfaces.

Justification

- **Independent Scalability:** Different parts of the courier system experience different workloads. For example, the Tracking and Notification Service may receive frequent location and delivery-status updates, while the Delivery Verification Service is mainly used when a delivery is completed. Microservices allow heavily used services to be scaled independently.
- **Fault Isolation:** The system contains services with clearly separated responsibilities. A failure in one service, such as the Route Optimization Service, can be isolated without necessarily bringing down the entire courier system.
- **Maintainability:** Each service has a well-defined responsibility. Route calculation, delivery verification, courier dispatch, and tracking can be developed, tested, and modified independently without affecting unrelated services.
- **Security:** Security-sensitive operations can be isolated within dedicated services. The Delivery Verification Service can handle authentication, authorization, input validation, and secure OTP verification. Communication between services can also be protected using authenticated and encrypted APIs.
- **Performance:** The Route Optimization Service can independently process route calculations, while the Tracking and Notification Service handles frequent tracking and delivery-status updates. This separation reduces the possibility of a single service becoming a performance bottleneck.
- **Flexibility and Deployment:** Individual services can be updated or deployed independently. Changes to one service do not require the entire application to be redeployed, making the system easier to evolve as courier requirements change.

Component Mapping

The major components identified in the system are:

- **Sender Client** – Allows the sender to initiate courier requests and receive delivery-related information.
- **Courier Dispatch Service** – Coordinates courier requests and manages communication with the supporting services.
- **Route Optimization Service** – Determines efficient routes for courier delivery operations.
- **Delivery Verification Service** – Handles delivery verification, including OTP-based verification.
- **Tracking and Notification Service** – Handles delivery tracking and sends status and notification updates.

Conclusion

Microservices Architecture provides modularity, independent scalability, fault isolation, improved maintainability, and better handling of security and performance-sensitive operations. Therefore, it is an appropriate architectural choice for the Hyperlocal Courier Dispatch and Tracking System.